

non-rel. Schrödinger equation
or relativistic Dirac equation

LDA or GGA
or hybrids

physical orbitals or not
mesh density and basis set

The diagram illustrates the Kohn-Sham equation and its components, with arrows indicating their physical origins:

$$\left(-\frac{\hbar^2}{2m} \nabla_{\vec{r}}^2 + v_{\text{ext}}(\vec{r}) + v_H(\vec{r}) + v_{xc} \right) \phi_i(\vec{r}) = E_i \phi_i(\vec{r})$$

Arrows point from the following descriptions to the corresponding terms in the equation:

- non-rel. Schrödinger equation or relativistic Dirac equation** points to the kinetic energy term $-\frac{\hbar^2}{2m} \nabla_{\vec{r}}^2$.
- pseudopotential (ultrasoft/PAW/norm-conserving) or all-electron** points to the external potential term $v_{\text{ext}}(\vec{r})$.
- LDA or GGA or hybrids** points to the exchange-correlation potential term v_{xc} .
- Hartree potential from solving Poisson eq. or integrating charge density** points to the Hartree potential term $v_H(\vec{r})$.
- physical orbitals or not** points to the orbital $\phi_i(\vec{r})$ on the left side of the equation.
- view EVs as mere Lagrange multipliers or band structure approx** points to the energy eigenvalue E_i .
- mesh density and basis set** points to the orbital $\phi_i(\vec{r})$ on the right side of the equation.